

Amjad S Althabteh

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Summary

C++ Graphics Engineer focused on high-performance rendering, low-latency systems, and engine architecture. Shipped real-time rendering engines, GPU-driven pipelines, and diagnostics tooling in C++17–C++23.

Education

B.S. in Computer Science — Wayne State University

Detroit, MI || GPA: 3.65/4.00

Experience

Graphics Engineer	Immersalab Graphics	Detroit, MI	Sep 2025 - Dec 2025
- Architected a C++20/OpenGL real-time rendering engine with GPU-driven pipelines, achieving ~25% improvement in stability - Built internal UI tools and a JSON asset pipeline supporting 10+ configurable scene parameters for rapid iteration - Implemented data-driven optimizations to balance GPU workloads, reducing GPU idle time by ~30% across test scenes - Developed GPU profiling and frame-timing tools, cutting render bottleneck diagnosis time by ~50%			
Indie Game Dev	Unreal C++ Developer	Ann Arbor, MI	Apr 2025 - Aug 2025
- Developed 8+ gameplay systems in C++20 and Unreal Blueprints integrating animation state machines and branching narrative logic - Produced a 90-second real-time cinematic using MetaHuman, Control Rig, and Sequencer — viewable in portfolio - Built animation control systems supporting smooth multi-layer blending across 15+ character animation states			
Software Development Intern	Rare Munchies	Ann Arbor, MI	Sep 2023 - Apr 2024
- Integrated UI components using TypeScript, improving page transition speed by ~20% and interaction responsiveness - Identified and resolved 8+ cross-browser compatibility issues, reducing debugging time by ~2hrs per sprint and unblocking delivery - Optimized client-side rendering logic and state management, reducing unexpected UI re-renders by ~35%			
Undergraduate Researcher — Orbital Collision Modeling			
Wayne State University — Detroit, MI Feb 2025 – Present			
- Modeled 50+ satellite trajectories using numerical integration and N-body gravitational dynamics - Simulated orbital collision probabilities across 1M+ trajectory samples using spatial partitioning, reducing compute time by ~40%			

Projects

Glint 3D – C++ 3D Rendering Engine	glint-rendering
- Optimized a C++17/OpenGL rendering engine, improving GPU workload distribution and reducing frame time by ~30%. - Implemented frustum culling and instanced rendering, cutting draw calls by ~45% in large scenes - Designed a custom memory pool allocator, reducing heap allocations by ~60% and runtime overhead	
3D Space Simulation Engine – C++ / OpenGL	space-simulations
- Developed a C++17/OpenGL engine supporting dynamic lighting across 10K+ simulated entities with stable frame delivery - Designed 4 modular engine systems (rendering, physics, math, entity) with clean separation of concerns - Implemented a custom math library (Vec3, Mat4) and camera system supporting full 3D projection/view pipeline.	
HFT OrderBook Engine — C++20	
- Implemented a price-time FIFO order matching engine processing ~2.1M orders/sec with sub-microsecond match latency - Designed cache-aligned bid/ask structures in C++20, reducing memory latency and eliminating false sharing	
Traceforge – C++ Runtime Stack Trace & Diagnostics Engine	
- Developed a C++23 diagnostics engine for stack trace parsing across 5+ build configs, cutting debug triage time by ~60% - Implemented memory tracking and automated regression generation across 10K+ test runs	

Skills

- Programming:** C++ (C++11–C++23), C, C#, Python, TypeScript, Java, SQL
- Graphics & Engines & Tools:** OpenGL 4.x, GLSL, Unreal Engine 4–5, Unity, Godot, CMake, VS Code, Git
- Systems & Performance:** RAII, custom allocators, memory pools, lock-free data structures, cache optimization, profiling (Valgrind, perf)